EXP NO: 8 EXPERIMENT TO UNDERSTAND LINEAR REGRESSION FOR A GIVEN DATA SET

AIM:

Understand the Linear regression for the dataset given.

DATASET GIVEN:

YearsExperience	Salary
1.1	39343
1.3	46205
1.5	37731
2	43525
2.2	39891
2.9	56642
3	60150
3.2	54445
3.2	64445
3.7	57189
3.9	63218
4	55794
4	56957
4.1	57081
4.5	61111
4.9	67938
5.1	66029
5.3	83088
5.9	81363
6	93940
6.8	91738
7.1	98273
7.9	101302
8.2	113812
8.7	109431
9	105582
9.5	116969
9.6	112635
10.3	122391
10.5	121872

PYTHON CODE:

```
import numpy as np
import pandas as pd
data = {
   'YearsExperience': [
      1.1, 1.3, 1.5, 2, 2.2, 2.9, 3, 3.2, 3.2, 3.7,
      3.9, 4, 4, 4.1, 4.5, 4.9, 5.1, 5.3, 5.9, 6,
      6.8, 7.1, 7.9, 8.2, 8.7, 9, 9.5, 9.6, 10.3, 10.5
   ],
   'Salary': [
      39343, 46205, 37731, 43525, 39891, 56642, 60150, 54445,
      64445, 57189, 63218, 55794, 56957, 57081, 61111, 67938,
      66029, 83088, 81363, 93940, 91738, 98273, 101302,
      113812, 109431, 105582, 116969, 112635, 122391, 121872
   ]
}
df = pd.DataFrame(data)
df.info()
df.dropna(inplace=True)
df.info()
df.describe()
features = df.iloc[:, [0]].values
label = df.iloc[:, [1]].values
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(
   features, label, test_size=0.2, random_state=42
)
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(x_train, y_train)
model.score(x_train, y_train)
model.score(x_test, y_test)
model.coef_
model.intercept_

import pickle
pickle.dump(model, open('SalaryPred.model', 'wb'))
model = pickle.load(open('SalaryPred.model', 'rb'))
yr_of_exp = float(input("Enter Years of Experience: "))
yr_of_exp_NP = np.array([[yr_of_exp]])
Salary = model.predict(yr_of_exp_NP)
print("Estimated Salary for {} years of experience is {}: ".format( yr_of_exp, Salary))
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
Enter Years of Experience: 7
Estimated Salary for 7.0 years of experience is [[91288.29027299]]:
```

RESULT:

Thus the code is executed successfully.